

Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from 3,900 transactions to guide strategic business decisions.



Dataset at a Glance

3,900

Purchases

18

Features

37

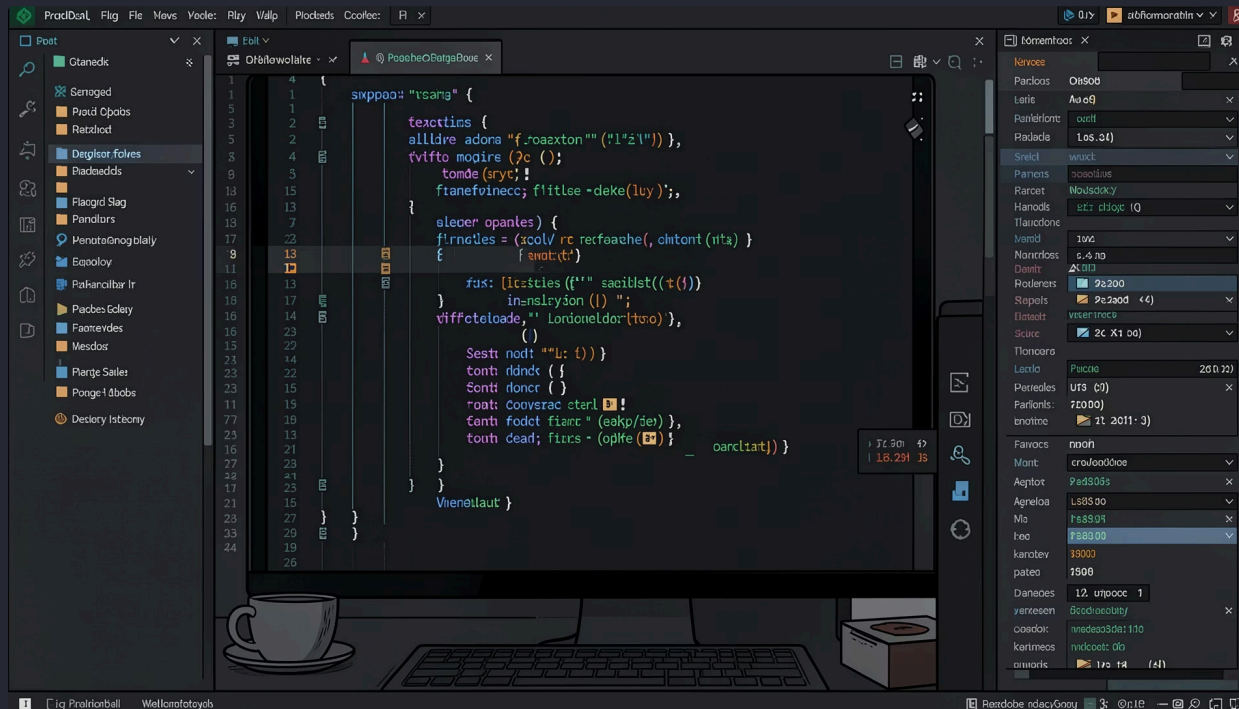
Missing Values

Review Rating column

Key features: demographics, purchase details, shopping behavior, and subscription status.



Data Preparation in Python



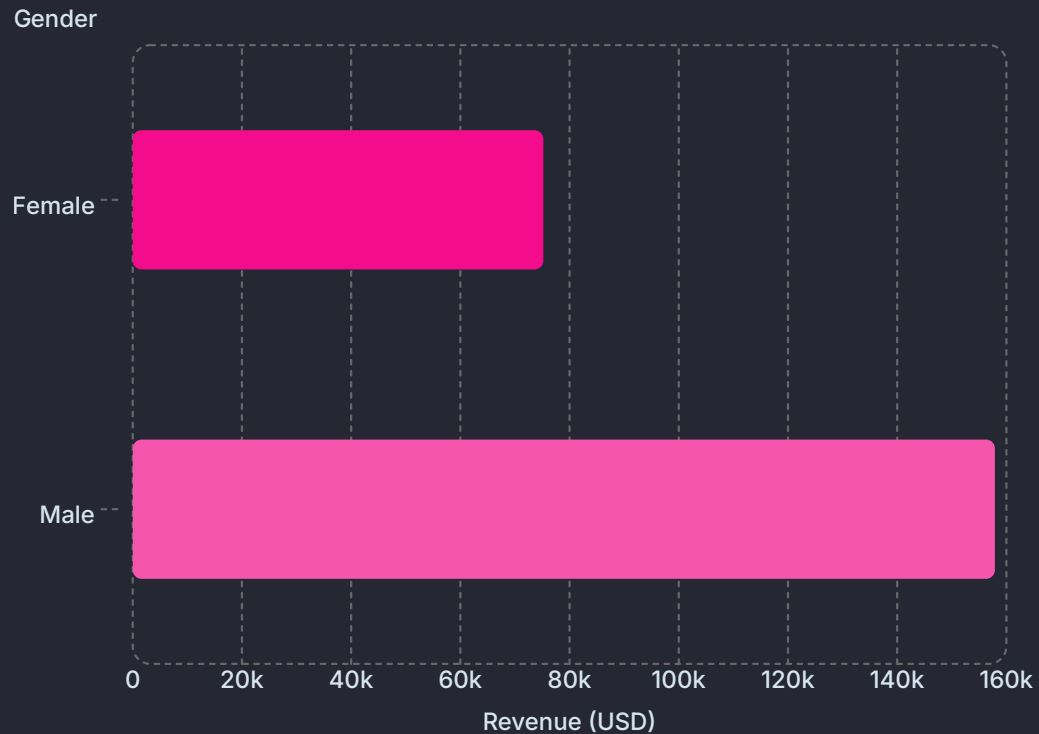
→ Load & Explore
pandas, df.info(), describe()

→ Clean
Imputed missing Review
Ratings by category median

→ Engineer
age_group bins,
purchase_frequency_days

→ Load
Cleaned DataFrame into
PostgreSQL

Revenue by Gender



Key Finding

Male customers generated **2.1x more revenue** than female customers — a significant gap worth investigating for targeted campaigns.

Top Products & Discount Insights

Top 5 by Rating

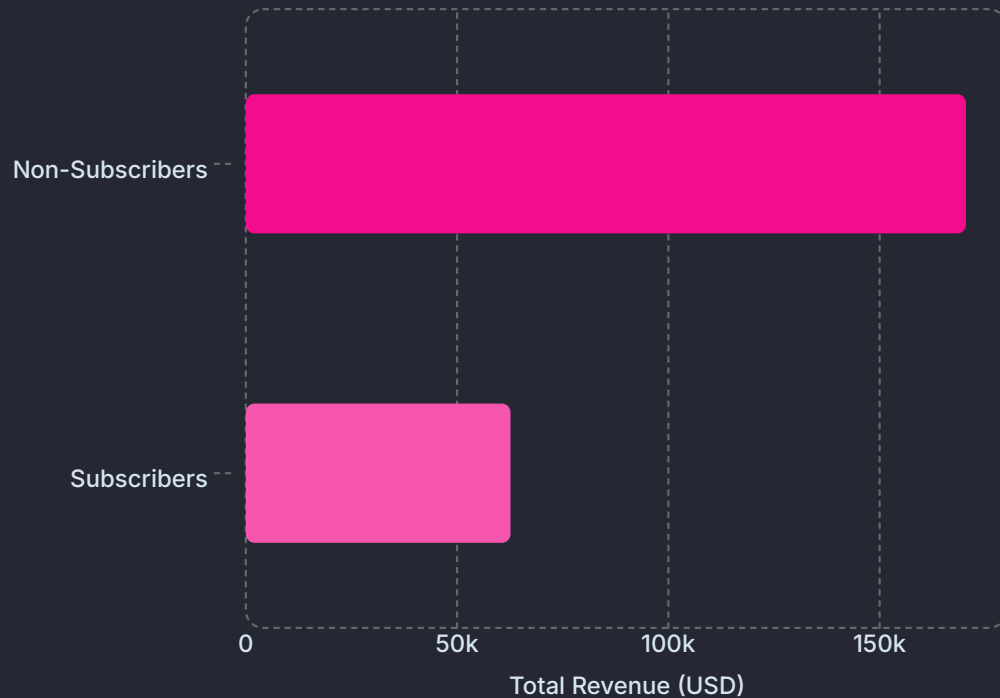
Item	Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discounted Products

Item	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Subscription & Shipping Behavior

Subscription



Key Findings

27% Subscribers

1,053 subscribers vs.
2,847 non-subscribers

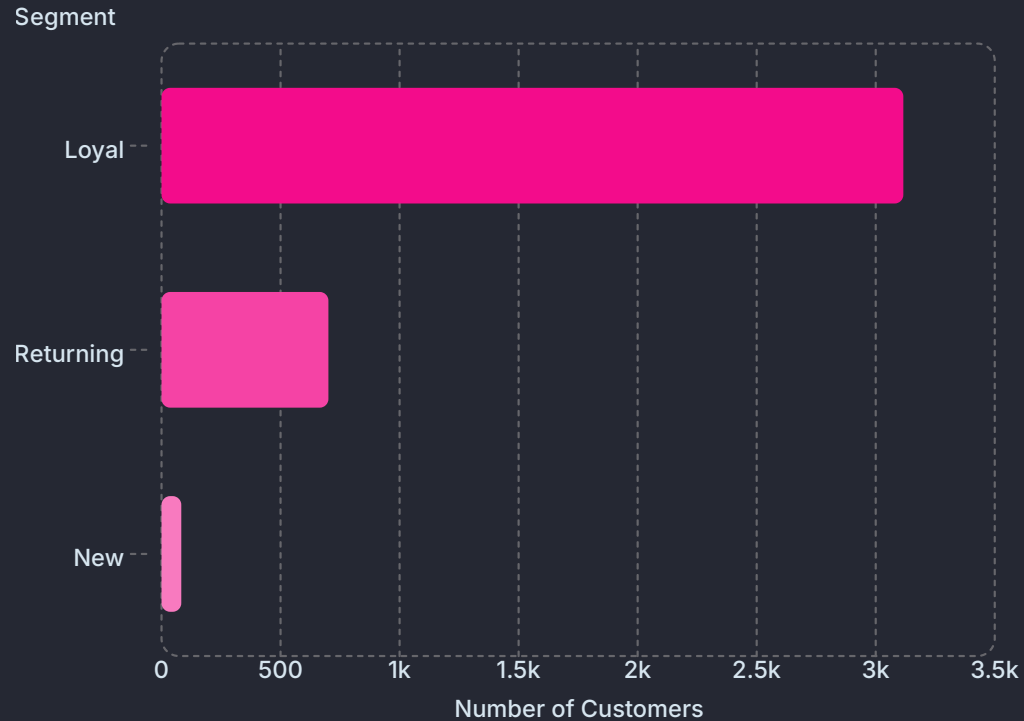
Similar Avg Spend

\$59.49 (sub) vs. \$59.87
(non-sub)

Express vs. Standard

\$60.48 vs. \$58.46 avg purchase

Customer Segmentation



Segment Breakdown

Loyal

3,116 customers — 80% of base

Returning

701 customers

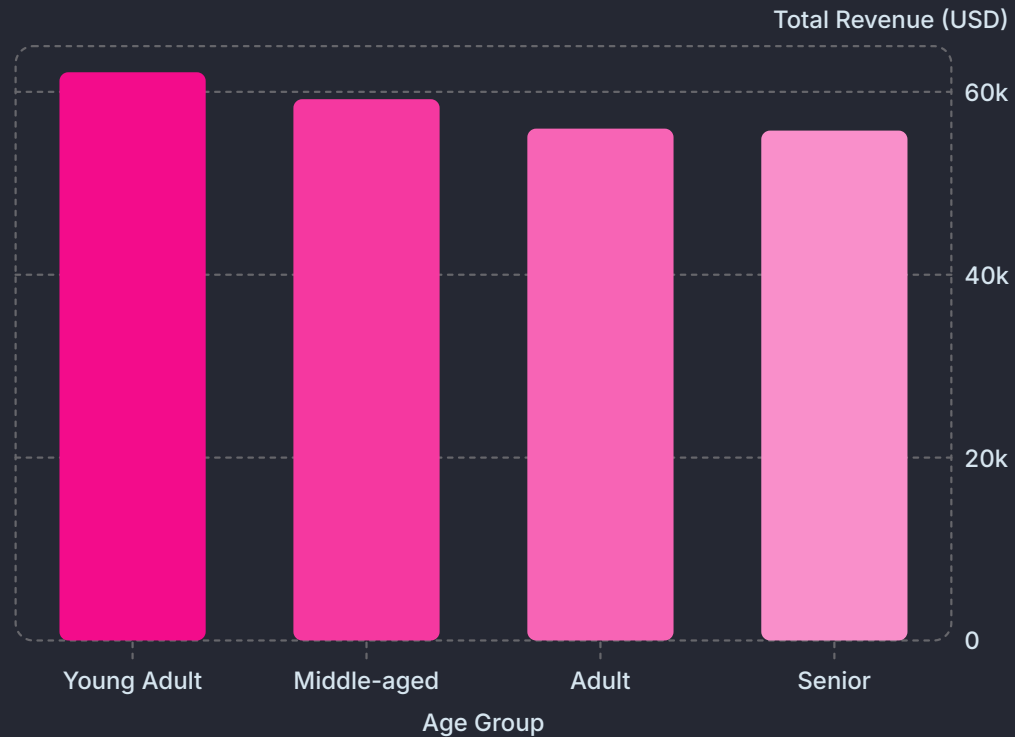
New

83 customers — growth opportunity



Repeat buyers: 2,518 non-subscribers vs. 958 subscribers

Revenue by Age Group



Insight

Young Adults lead in revenue contribution. Revenue declines gradually with age — a clear signal for age-targeted marketing strategies.

Power BI Dashboard

3.9K

Customers tracked

\$59.76

Avg purchase amount

3.75

Avg review rating

27%

Subscriber share



